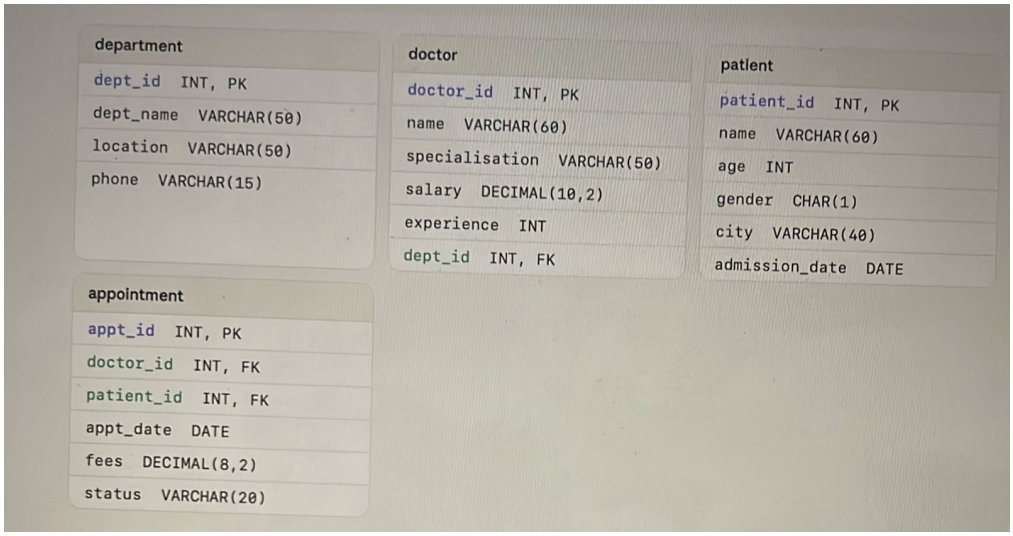

department

dept_id	INT, PK
dept_name	VARCHAR(50)
location	VARCHAR(50)
phone	VARCHAR(15)

doctor

doctor_id	INT, PK
name	VARCHAR(60)
specialisation	VARCHAR(50)
salary	DECIMAL(10,2)
experience	INT
dept_id	INT, FK

patient

patient_id	INT, PK
name	VARCHAR(60)
age	INT
gender	CHAR(1)
city	VARCHAR(40)
admission_date	DATE

appointment

appt_id	INT, PK
doctor_id	INT, FK
patient_id	INT, FK
appt_date	DATE
fees	DECIMAL(8,2)
status	VARCHAR(20)

Q1 — CREATE TABLE WITH CONSTRAINTS

1. Create the department table. Apply PRIMARY KEY on dept_id and a UNIQUE constraint on dept_name.
2. Create the doctor table. Apply PRIMARY KEY, NOT NULL on doctor_id, a CHECK constraint to ensure salary > 0, and a FOREIGN KEY referencing department.
3. Create the patient table. Apply PRIMARY KEY, NOT NULL on patient_id, and a CHECK constraint so gender can only be 'M' or 'F'. Set admission_date to default to today's date using DEFAULT CURDATE().
4. Create the appointment table. Apply PRIMARY KEY, two FOREIGN KEY constraints referencing doctorand patient, and a CHECK so fees >= 0. Set default status to 'Scheduled'.
5. Insert at least 5 records into each of the four tables. Make sure the data is realistic — include doctors from at least 3 departments, patients from different cities, and a mix of appointment statuses ('Scheduled', 'Completed', 'Cancelled').

Q2

1. Add a new column email of type VARCHAR(80) to the doctor table.
2. Add a new column blood_group of type VARCHAR(5) to the patient table.
3. Change the data type of phone in the department table from VARCHAR(15) to VARCHAR(20).
4. Rename the column status to appt_status in the appointment table.
5. Add a NOT NULL constraint to the specialisation column of the doctor table.
6. Drop the location column from the department table.

Q3 —

1. Display the names and salaries of all doctors whose salary is greater than 60000, ordered by salary in descending order.
2. List all patients whose age is between 30 and 60 (inclusive) and whose city is either 'Mumbai' or 'Delhi'.
3. Find all doctors whose name starts with the letter 'A' using the LIKE operator.
4. Display all appointments where the appt_status is 'Completed' and fees are greater than 500.
5. Show all patients whose blood_group is NULL (i.e., not yet recorded).
6. Count the total number of patients admitted in the year 2024 using YEAR(admission_date).
7. Find the maximum, minimum, and average fees from the appointment table.
8. Display the specialisation and the count of doctors in each specialisation. Show only those specialisations that have more than 1 doctor. Use GROUP BY and HAVING.

Q4.

1. Using INNER JOIN, display each doctor's name along with the department name they belong to.
2. Using LEFT JOIN, display all departments and the names of doctors in each department. Include departments that currently have no doctors.
3. Display the patient name, doctor name, appointment date, and fees for every completed appointment.

4. Find the names of patients who have **never** had an appointment.
5. Display the department name and the total fees collected from all appointments in that department.